Signed words and polynomial EEA: independent written review

Source-first mathematical review

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Both published derivations are supported as conditional written arguments. No correction to their principal equations was found. This review does not certify a newly executed Lean proof or native EEA implementation.

Signed words. Signs in $\{1, -1\}$ preserve the literal i8 operations and make the same-root denominator exactly ± 2. Canonical addition fits u64 at $r \leq 2^{63}$. This bound is explicitly present at the cited fixture, not implied by the public Ring constructor. For $r = 2h + 1$, inverse2 is $h + 1$ and inverse(-2) is h. The invalid-sign example at101 correctly installs50 instead of1, demonstrating the necessary sign premise.

Six-register EEA. Literal native updates agree with the simultaneous mathematical recurrence. Both Bezout equalities and common-divisor equivalence are preserved. Each active step decreases the second remainder's length; $b.length$ bounds active steps. A future fuel API must separately specify whether its terminal observation consumes fuel. Scaling all three final outputs by the same nonzero inverse leading coefficient gives the normalized endpoint.

Fallback exactness. The two EEA endpoints give $D \mid U_a, U_b, V_a + V_b$. Therefore $D^2 \mid U_a U_b$. For the numerator, use

$$F + V_a V_b = (F - V_a^2) + V_a(V_a + V_b).$$

Every term is divisible by D, including shared support and inverse pairs. This proves the two exact-division implications; the subsequent Mumford composition identity remains separate.

Fixture status. Hand polynomial expansion confirms the proposed F17 Bezout coefficients and the degree-one inverse/doubling examples. Their native algorithm executions remain not executed. No new measurement or cohort result is supplied.

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